SHASHANK CHIKATTIMALA

Senior Software Engineer | AI / LLMOps Engineer & Architect
Charlotte, NC | shashankuidev9@gmail.com | +1 248-919-8701 | linkedin.com/in/shashankcm90 | github.com/shashankcm

PROFESSIONAL SUMMARY

Senior engineering leader who has shipped enterprise systems for Lowe's, Walmart, Bridgewater and the State of Arkansas. Currently lead 8 engineers and own architecture, release quality and the Docker/Kubernetes/CI-CD plumbing underneath.

Focused on production AI: Enterprise Agentic RAG, vLLM / TensorRT-LLM serving, AWQ/GGUF quantization, and the guardrails and evaluation that keep LLM features from breaking in production.

Care about latency, cost per token, and systems that fail loudly. Mentor engineers, set standards, make the build-vs-buy calls.

TECHNICAL SKILLS

- AI Orchestration & Frameworks: LangGraph, LlamaIndex, LangChain, Enterprise Agentic RAG, multi-agent routing
- LLMs & Foundations: Claude 3.5 Sonnet, GPT-4o, Llama-3 (70B), Hugging Face, PyTorch
- AI Safety, Guardrails & Evaluation: NeMo Guardrails, Guardrails AI, TruLens, LangSmith, Ragas
- MLOps / LLMOps / Compute: vLLM, TensorRT-LLM, Triton Inference Server, AWQ, GGUF, Distributed Inference, DeepSpeed, Megatron-LM, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Azure DevOps, Octopus Deploy, Ansible, Gradle, Bash, Prometheus, Grafana, Hybrid Cloud / On-Prem
- Data & Vector Databases: Qdrant, Milvus, Redis, MongoDB, SQL, DB2, Cosmos DB, MinIO
- Core Languages & Backend: Python, TypeScript, JavaScript, Java, XML, FastAPI, Node.js, Express, Swagger, Apollo / URQL, WebSockets, gRPC, Celery, REST, GraphQL
- Cloud & Frontend: AWS (Lambda, SQS, SNS, Kinesis, S3, EC2, RDS, DynamoDB, IAM, CloudWatch, CloudFormation, CloudFront, Route 53), Azure (AKS, Functions, Data Factory, Blob Storage, VMs), React, Angular / AngularJS, Next.js, SASS, Tailwind, Material UI, Storybook, Webpack, Babel, PWA, Redux, Redux Toolkit, Zustand, Module Federation
- Testing & Tooling: Jest, Enzyme, React Testing Library, Mocha / Sinon, Cypress, K6, Sauce Labs, Git, Bitbucket, Jira, Confluence, Splunk, Postman, Agile / Scrum
- Leadership & Soft Skills: Technical leadership, mentoring, cross-functional communication, stakeholder alignment, code review, problem solving, architecture decisions

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Lowe's, Charlotte, NC | 06/2023 - Present

- **Team and delivery:** Lead 8 engineers building the pre- and post-fulfillment order management platform (React, Next.js). Communicate daily with product managers, UX designers and upstream/downstream teams to hold delivery dates and resolve blockers. User engagement up 40%.
- **Monolith to micro frontends:** One monolith meant every team shipped on the same schedule. Moved it to Webpack Module Federation with runtime loading and a versioning strategy that keeps rollouts backward compatible. Teams now ship independently.
- **Page load:** Next.js ISR, static optimization and dynamic imports gave 25% faster loads. Code-splitting with Suspense took another 20% off initial load.
- **Quality gates:** Set up Jest, React Testing Library, Cypress and K6 performance thresholds. Production bugs down 20%. Automated linting and deploys cut the release cycle 30%.

- **State and process:** Adopted Redux for complex state and built reusable components with hooks and Context API. Joined CAB meetings, sprint planning and retrospectives, flagged technical debt, and ran stage-wise releases including in-store dark-mode testing.
- **Leadership and mentoring:** Set front-end standards, ran code reviews, pair sessions and weekly learning sessions, mentored junior engineers, and contributed to the shared Lowe's UI component library. Standardized Docker environments and Jenkins pipelines with DevOps.

Sr. Full Stack Developer | Walmart (via EPAM Systems), Bentonville, AR | 11/2020 - 05/2023

- **Release engineering:** Build and release processes differed across divisions. Standardized them on Azure DevOps, Octopus Deploy, Ansible, PowerShell, Bash and Gradle, with scripted DEV/QA/UAT/PROD deployments.
- **Kubernetes and IaC:** Built AKS and self-hosted Kubernetes clusters with Terraform and Ansible. Dockerized test environments. Designed Azure Data Factory pipelines for data ingestion and publishing, and managed VMs and Blob Storage.
- **Applications:** Built a role-based React app (Context API, Redux) and Node/Express services. Tested with Jest, Mocha, Sauce Labs. Debugged production with Splunk.

Sr. Front End Developer | Bridgewater Associates (via EPAM Systems), Norwalk, CT | 04/2020 - 10/2020

- **Search for an investment firm:** Built a tokenized search (React, TypeScript, CodeMirror) with fuzzy matching and a debounced hook. IndexedDB caching skipped repeat database calls.
- **Quality and delivery:** Ag-Grid and Highcharts for data display, BackstopJS visual regression, CI/CD on AWS (EC2, S3, RDS, DynamoDB).

Full Stack Developer | Walmart (via eSystems Inc), Bentonville, AR | 10/2019 - 03/2020

- Built the associate preboarding app (React, Material-UI, Node, Azure Functions, Cosmos DB) so new hires submit details before day one.

Full Stack Developer | State of Arkansas DHS (via eSystems Inc), Little Rock, AR | 06/2017 - 10/2019

- **Configurable intake:** Built a drag-and-drop form builder in React (MongoDB templates, MySQL referrals) so power users change forms without a deploy.
- **Serverless and offline:** Node APIs on AWS Lambda with Cognito, SNS, SQS and Kinesis. PWA with service workers, redux-persist, web push notifications and IndexedDB works on dead networks.
- **Curam:** Customized IBM Curam (Java, XML, DB2, WebSphere) for healthcare eligibility, including the DHS S.A.V.E. immigration check. Built React/Redux portals and a React Native proof of concept.
- **Incident and accident reporting:** Built a React/Redux incident report system with reusable form components and background file uploads, on Node/Express REST services with MongoDB (Mongoose).

Full Stack Developer | Peta Digit Inc., India | 05/2011 - 07/2015

- Built an e-commerce site, a role-based CMS and a feedback dashboard (AngularJS, Node.js, jQuery / jQuery UI, JSP, SASS), working directly with business users and UX.

AI PROJECTS (GITHUB)

- **Enterprise Multi-Agent Swarm** (Python, LangGraph, LlamaIndex, Qdrant, Claude 3.5 Sonnet, FastAPI, Redis, Docker): LangGraph router splits queries into parallel vector and SQL calls. Ragas-based self-correction loop rewrites queries below 85% confidence, cutting hallucinations 42%. Redis-backed shared state cut context token overhead 30%.
- **High-Throughput Inference & LLMOps Engine** (vLLM, TensorRT-LLM, AWQ/GGUF, Llama-3-70B, Triton, Kubernetes, Prometheus): Served quantized 70B models across multi-GPU nodes with tuned tensor parallelism. 4-bit AWQ cut KV cache memory 60% and raised concurrency 3.5x. Continuous batching hit 2,400 tokens/sec per node, tracked in Grafana.

- **Agentic RAG with Hybrid Search** (Qdrant, Milvus, Cohere Rerank, PyTorch, MinIO, gRPC): Parent-child chunking, BM25 plus dense retrieval and Cohere reranking lifted Top-10 precision 34%. Cross-encoder context compression cut 180ms of LLM latency.
- **Code Generation & Execution Sandbox** (Python, GPT-4o, Docker, FastAPI, Celery, Redis, WebSocket): AST guardrails and timeouts gave 94% safe execution on HumanEval. Worker pool runs 50 isolated environments with sub-1.2s response.
- **CI LLM Guardrail & Evaluation Engine** (GitHub Actions, NeMo Guardrails, TruLens, vLLM, Prometheus): Pipeline gates fine-tuned models on safety, toxicity and drift before staging. NeMo blocks prompt injection under 15ms. TruLens and Prometheus alert on live drift.
- **Distributed Inference & Training Infrastructure** (vLLM, TensorRT-LLM, DeepSpeed, Megatron-LM, Kubernetes): Tuned vLLM paged attention, cutting peak VRAM from about 78 GB to 54 GB per GPU at 32k context on 8x A100. AWQ 4-bit gave 2.6x FP16 tokens/sec at under 0.3 perplexity increase. DeepSpeed/Megatron-LM fine-tune scaled to about 85% efficiency at 8 GPUs.

EDUCATION & CERTIFICATIONS

- M.S., Computer Science, University of Michigan-Flint, 2017
- B.Tech, Information Technology, K.S. Raju Institute of Technology and Sciences, JNTU, 2011
- IBM Curam Certified Application Developer v7, 2018